PYROBIX PROJECT · 2026

VISION-BASED ROBOTIC SORTING

ABB IRB 1200 · MITSUBISHI FX5U · SICK InspectorP · DELTA VFD Conveyor

ABB IRB 1200 | Mitsubishi FX5U | SICK InspectorP | DELTA VFD Conveyor

● ROBOT ONLINE ● VISION DETECTING ● CONVEYOR RUNNING ● PLC AUTO CYCLE 4.2s SORTED 1,247

01 – PROCESS FLOW

PART ENTRY
VFD Conveyor
START signal

▶

VISION
SICK InspectorP
Color + Position

▶

PLC LOGIC
Mitsubishi FX5U
Sort Decision

▶

ROBOT PICK
ABB IRB 1200
RAPID Program

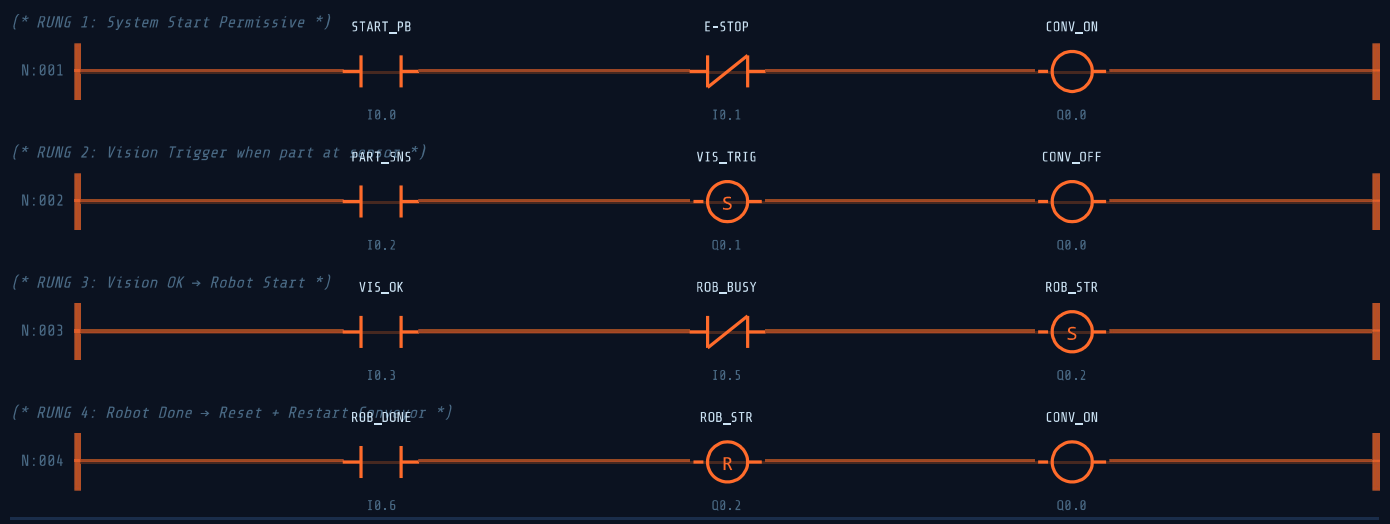
▶

SORT & PLACE
3 Bins by Color
Cycle Complete

02 – PLC LADDER DIAGRAM

PLC LADDER LOGIC

IEC 61131-3 · LD



KEY: —| |— NO Contact | —|/|— NC Contact | —( )— Coil | —(S)— Set | —(R)— Reset

ABB RAPID PROGRAM

RAPID · IRB 1200

```
! MODULE: SortingSystem · ABB IRB 1200 · RAPID v6
MODULE SortingSystem

  PERS tooldata  tGripper := [TRUE,[[0,0,120],[1,0,0,0]],...];
  CONST robtarget pBin_RED   := [[300,-150,80],...];
  CONST robtarget pBin_BLUE  := [[300,0,80],...];
  CONST robtarget pBin_GREEN := [[300,150,80],...];
  CONST robtarget pHome      := [[0,0,400],...];
  VAR   num       nColor     := 0;
  VAR   robtarget pPick;

PROC main()
  MoveJ pHome, v400, z50, tGripper;
```

```
  WHILE TRUE DO
    WaitDI di_RobStart, 1;        ! Wait PLC start
    SetDO  do_RobBusy, 1;
    nColor := GInput(giColorCode);
    pPick  := ReadVisionPos();
    MoveJ Offs(pPick,0,0,100), v600, z10, tGripper;
    MoveL pPick, v80, fine, tGripper;
    SetDO do_GripClose, 1;
    WaitTime 0.3;
    MoveL Offs(pPick,0,0,150), v400, z20, tGripper;
    SortToBin nColor;
    SetDO do_RobBusy, 0;
    SetDO do_RobDone, 1;
    WaitTime 0.1;
    SetDO do_RobDone, 0;
  ENDWHILE
ENDPROC

PROC SortToBin(num id)
  VAR robtarget pT;
  TEST id
    CASE 1: pT:=pBin_RED;
    CASE 2: pT:=pBin_BLUE;
    CASE 3: pT:=pBin_GREEN;
    DEFAULT: ErrLog "Unknown color!";
  ENDTEST
  MoveJ Offs(pT,0,0,120), v600, z10, tGripper;
  MoveL pT, v100, fine, tGripper;
  SetDO do_GripOpen, 1;
  WaitTime 0.2;
  MoveL Offs(pT,0,0,120), v400, z20, tGripper;
  MoveJ pHome, v600, z50, tGripper;
ENDPROC

ENDMODULE
```

03 — I/O SIGNAL MAP

PLC INPUTS

START_PB	I0.0
E-STOP	I0.1
PART_SENSOR	I0.2
VISION_OK	I0.3
COLOR[2:0]	I0.4-6
ROB_BUSY	I0.5
ROB_DONE	I0.6

PLC OUTPUTS

CONV_START	Q0.0
VIS_TRIGGER	Q0.1
ROB_START	Q0.2
STATUS_LAMP	Q0.3
ALARM_OUT	Q0.4
SORT_BIN[2:0]	Q0.5-7

COMMS

PLC ↔ ABB	DeviceNet
PLC ↔ SICK	Ethernet/IP
PLC ↔ VFD	Modbus RTU
Vision → PLC	4×DI Color
Vision → ABB	X/Y via DSQC
Speed Ref	0-10V AO

SYS REV 2.4 | IRB1200-7/0.7 | FX5U-32MR | InspectorP611
RAPID v6 · IEC 61131-3 LD · DeviceNet · Ethernet/IP

#Automation #Robotics #PLC #MachineVision #Industry40 #ABB #SmartFactory #PYROBIXProject